

Brenda Iradukunda

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EDUCATION

Washington and Lee University,

2027 | Lexington, VA

Bachelor of Science in Computer Science and Bachelor of Arts in Economics, GPA: 3.63/4.0

Relevant Courses: Software Development, Data Structures, Computer Organization, Linear Algebra, Discrete Math, Programming Language Design, Algorithm Design and Analysis, Web Development

PROFESSIONAL EXPERIENCE

Washington and Lee University,, *Work Study - Global Discovery Lab Specialist*

08/2024 – Present | Lexington, VA

- Maintain and customize web content for the Global Discovery Lab and Virginia Governor's World Language Academies, improving site performance by 25% and user engagement by 18%
- Automate placement testing workflows and data processing, reducing manual effort by 40% and improving assessment accuracy by 20%
- Train students and staff to access and navigate language-learning platforms (Mango Languages, Sanako), increasing adoption and usage by 35%

KLab Innovation Hub, *Software Engineering Intern*

06/2025 – 08/2025 | Kigali, Rwanda

- Developed web modules for the Future Koders platform using Flask and JavaScript, improving accessibility and enhancing user interaction by 30%
- Optimized backend APIs and 7 database queries for the Digital Skills for Employment program, enhancing system performance and reliability by 35%
- Collaborated with senior engineers on code reviews and debugging, reinforcing best practices in testing and scalable backend development

Washington and Lee University, *Peer Tutor*

09/2024 – 05/2025 | Lexington, VA

- Delivered 75+ hours of targeted 1:1 and group tutoring, leveraging live coding and data visualization to demystify recursion, graph algorithms, and algorithmic complexity, boosting student assignment performance
- Guided peers through structured debugging workflows and code optimization techniques, achieving a 20% reduction in project defects

TECHNICAL PROJECTS & VOLUNTEERING

Picasso Expression Image Generator, *Java, Swing, JUnit, Git*

10/2025 – 12/2025

- Built a Java Swing application that parses mathematical expressions into procedurally generated images using a tokenizer, parser, expression tree, and evaluation pipeline
- Implemented binary operators and precedence handling, contributing to 86% JUnit test coverage across 16K+ instructions
- Developed a history feature enabling keyboard navigation through previously evaluated expressions
- Collaborated in a 5-person Git team using feature branches to debug parser logic and integrate GUI functionality

EduTrack, *Flask, Python, PostgreSQL, JavaScript, HTML, CSS*

08/2025 – 10/2025

- Developed a data-driven web application with Flask and PostgreSQL to monitor student progress and academic outcomes
- Implemented interactive dashboards to visualize academic trends and generate real-time progress insights
- Built responsive user interactions and dynamic content updates with JavaScript and Flask to enhance usability

Voice Library, *Flask, Python, SpeechRecognition, pyttsx3, HTML, CSS, JavaScript*

06/2025 – 07/2025

- Developed a Flask-based web app that lets users search for books via text or voice, improving accessibility and interaction by 40%
- Integrated SpeechRecognition and pyttsx3 libraries to enable real-time voice commands and spoken feedback
- Designed a responsive frontend with HTML, CSS, and JavaScript for a seamless user experience across input modes

Women in Technology and Science, *Facilitator*

09/2023 – 05/2025

- Designed and delivered programming workshops for high school students, tailoring curriculum based on feedback and learning outcomes
- Mentored 15 students in building and deploying personal websites, emphasizing usability and real-world problem-solving

SKILLS & TOOLS

Languages: Java, Python, C++, C, JavaScript, SQL, HTML/CSS, R, Bash

Frameworks/Tools: React, Spring, Flask, Node.js, REST APIs, Eclipse, LaTeX, Linux, VS code, Azure

Developer Tools: Git, GitHub

Professional Skills: Leadership, Mentorship, Cross-functional Communication, Tutoring

PROGRAMS & AFFILIATIONS

CodePath, ColorStack, Rewriting the Code, Girls Who Code, Cybersecurity club, Grace Hopper Celebration' 25, @Scale: Product 2025 (Meta)